

We now explain how many concepts from the theory of schemes may be applied to algebraic stacks.

Let  $\mathbf{P}$  be a property of morphisms of schemes, stable by étale change of base, and of a local nature (for the étale topology) on the target.

For instance: being an open immersion with dense image, being dominant, birational...

A representable morphism of algebraic stacks  $f: T_1 \rightarrow T_2$ , is said to have property  $\mathbf{P}$  if for one (and hence for every) surjective étale morphism  $x: X \rightarrow T_2$ , the morphism of schemes deduced by base change  $f': X \times_{T_2} T_1 \rightarrow X$  has that property.

Let  $\mathbf{P}$  be a property of morphisms of schemes, which, at source and target, is of a local nature for the étale topology. This means that, for any family of commutative squares

$$\begin{array}{ccc} X_i & \xrightarrow{g_i} & X \\ \downarrow f_i & & \downarrow f \\ Y_i & \xrightarrow{h_i} & Y \end{array}$$

where the  $g_i$  (resp.  $h_i$ ) are étale and cover  $X$  (resp.  $Y$ ):

$$\mathbf{P}(f) \Leftrightarrow \forall_i \mathbf{P}(f_i).$$

For instance:  $f$  flat, smooth, étale, unramified, normal, locally of finite type, locally of finite presentation.

If  $f: T_1 \rightarrow T_2$  is a morphism of algebraic stacks, we say  $f$  has property  $\mathbf{P}$  if for one, then necessarily for every, commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{x} & T_1 \\ \downarrow f' & & \downarrow f \\ Y & \xrightarrow{y} & T_2 \end{array}$$

where  $X$  and  $Y$  are schemes and  $x, y$  are étale and surjective,  $f'$  has property  $\mathbf{P}$ .

Similarly, if  $\mathbf{P}$  is a property of schemes, of a local nature for the étale topology, an algebraic stack  $T$  will be said to have property  $\mathbf{P}$  if for one (and hence for every) surjective étale morphism  $x: X \rightarrow T$ ,  $X$  has property  $\mathbf{P}$ . This applies to, for instance, the properties of being regular, normal, locally noetherian, of characteristic  $p$ , reduced, Cohen-Macaulay...

An algebraic stack  $T$  will be called *quasi-compact* if there exists a surjective étale morphism  $x: X \rightarrow T$  with  $X$  quasi-compact. A morphism  $f: T_1 \rightarrow T_2$  of algebraic stacks will be called *quasi-compact* if for any quasi-compact scheme  $X$  over  $T_2$ , the fiber product  $T_1 \times_{T_2} X$  is quasi-compact. It is enough to test the condition for a surjective family  $f_i: X_i \rightarrow T_2$ . We define a morphism  $f: T_1 \rightarrow T_2$  to be of *finite type*, if it is quasi-compact, and locally of finite type; of *finite presentation*, if it is quasi-compact, quasi-

separated, and locally of finite presentation. An algebraic stack is *noetherian*, if it is quasi-compact, quasi-separated, and locally noetherian.

The key point in what follows will be the definition of a "proper morphism" and the analogue of Chow's lemma.

A morphism  $f: T_1 \rightarrow T_2$  is said to have a property  $P$ , locally on  $T_2$ , if there exists a surjective étale morphism  $x: X \rightarrow T_2$  such that the morphism  $f'$  deduced from  $f$  by the change of base by  $x$  has property  $P$ .

**Definition (4.11).** — A morphism  $f: T_1 \rightarrow T_2$  is proper if it is separated, of finite type and if, locally over  $T_2$ , there exists commutative diagrams

$$\begin{array}{ccc} & T_3 & \\ g \swarrow & & \searrow h \\ T_1 & & T_2 \end{array}$$

with  $g$  surjective and  $h$  representable and proper.

The following form of Chow's Lemma will be sufficient for our purposes.

**Theorem (4.12).** — Let  $S$  be a noetherian scheme and  $f$  be a morphism from an étale site  $T$  to  $S$ . We assume  $f$  to be separated and of finite type. Then, there exists a commutative diagram

$$\begin{array}{ccccc} T & \xleftarrow{g} & T' & \xrightarrow{j} & T'' \\ \downarrow f & & \downarrow f' & & \downarrow f'' \\ S & & S & & S \end{array}$$

in which  $T'$  and  $T''$  are schemes and such that

- (i)  $g$  is proper, surjective and generically finite;
- (ii)  $j$  is an open immersion;
- (iii)  $f''$  is projective.

Using (4.12), it is easy to extend the cohomological theory of coherent sheaves to the present situation. In fact if  $f: T_1 \rightarrow T_2$  is a proper morphism of noetherian algebraic stacks and if  $\mathcal{F}$  is a coherent sheaf on  $T_1$ , then the  $R^i f_*(\mathcal{F})$  are coherent sheaves on  $T_2$ .

However, the  $R^i f_*(\mathcal{F})$  don't need to be zero for  $i$  large enough. Let  $S$  be a scheme and  $G$  a finite group. We denote by  $p$  the projection  $p: [S/G] \rightarrow S$ . Quasi-coherent (resp. coherent) sheaves of modules on  $[S/G]$  may (and will) be identified with quasi-coherent (resp. coherent) sheaves of modules on  $S$ , on which  $G$  acts. One has

$$R^i p_*(\mathcal{F}) \cong H^i(G, \mathcal{F}).$$

In general, the (quasi-coherent sheaf) cohomology of algebraic stacks appears as a mixture of finite group cohomology and of scheme cohomology.

The *disjoint sum*  $T$  of a family  $(T_i)_{i \in I}$  of stacks is the stack a section of which over a scheme  $X$  consists of

- (i) a decomposition  $X = \coprod_i X_i$  of  $X$ ;
- (ii) a section of  $T_i$  over  $X_i$  for each  $i$ .

The void stack  $\emptyset$  is the one represented by the void scheme.

A stack is *connected* if it is non-void and is not the disjoint sum of two non-void stacks.

**Proposition (4.13).** — *A locally noetherian algebraic stack is in one and only one way the disjoint sum of a family of connected algebraic stacks (called its connected components).*

We denote by  $\pi_0(T)$  the set of connected components of locally noetherian algebraic stack  $T$ . If  $x: X \rightarrow T$  is surjective and étale,  $\pi_0(T)$  is the cokernel of the two maps

$$\pi_0(X \times_T X) \rightrightarrows \pi_0(X) \rightarrow \pi_0(T).$$

**Proposition (4.14).** — *Let  $T$  be an algebraic stack of finite type over a field  $k$ . Then,  $T$  is connected if and only if there exists a connected scheme  $X$ , of finite type over  $k$ , and a surjective morphism from  $X$  to  $T$ .*

An open subset  $U$  of an algebraic stack  $T$  is a full subcategory  $U \subset T$  which is an algebraic stack, which contains together with any  $t \in \text{Ob}(T)$  all isomorphic  $t'$  and such that the inclusion  $j: U \rightarrow T$  is representable by open immersions. The open subsets of  $T$  corresponds bijectively to the open subsets of its étale site.

For each open subset  $U$  of  $T$ , there exists one and only one full subcategory  $T-U$  of  $T$ , which is an algebraic stack, which contains together with any  $t \in \text{Ob}(T)$  all isomorphic  $t'$  and such that

- (i)  $T-U$  is reduced;
- (ii) the inclusion map  $i: T-U \rightarrow T$  is representable by closed immersions;
- (iii) for any étale surjective morphism  $x: X \rightarrow T$ , the inverse image of  $T-U$  on  $X$  is the complement of the inverse image of  $U$ .

An algebraic stack  $F$  in  $T$  satisfying (i) and (ii) is a *closed subset* of  $T$ , and the functor  $U \mapsto T-U$  is an isomorphism of the set of open and the set of closed subsets of  $T$ . If  $F$  satisfies only (ii),  $F_{\text{red}}$  satisfies (i) and (ii) so that  $F$  defines a closed subset of  $T$ .

An algebraic stack  $T$  is *irreducible* if it is not the union of two closed subsets, non void and distinct from  $T$ .

**Proposition (4.15).** — *A noetherian algebraic stack  $T$  is in one and only one way the union of irreducible closed subsets, none of which contains any other. They are called the irreducible components of  $T$ . If  $U$  is an open dense subset of  $T$ , the irreducible components of  $U$  are the non-void intersections of  $U$  with the irreducible components of  $T$ .*

Each irreducible component of  $T$  is contained in a connected component of  $T$ . Conversely:

**Proposition (4.16).** — *The connected components of a normal noetherian algebraic stack are irreducible.*

**Theorem (4.17).** — *Let  $f$  be a morphism of finite type from an algebraic stack  $T$  to a noetherian scheme  $S$ . For  $s \in S$ , let  $n(s)$  be the number of connected components of the geometric fibre of  $T$  at  $s$ . Then*

- (i)  $n(s)$  is a constructible function of  $s$ ;

- (ii) if  $f$  is proper and flat, then  $n(s)$  is lower-semi-continuous;  
 (iii) if  $f$  is proper flat, and has geometrically normal fibres, then  $n(s)$  is constant.

Let  $f: T \rightarrow S$  be a morphism of finite type from an algebraic stack  $T$  to a noetherian scheme  $S$ . Assume that the diagonal map  $T \rightarrow T \times_S T$  is separated and quasi-compact.

**Theorem (4.18)** (Valuative criterion for separation.) — *The morphism  $f$  is separated if and only if, for any complete discrete valuation ring with algebraically closed residue field and any commutative diagram*

$$\begin{array}{ccc} & & T \\ g_1 \nearrow & & \downarrow f \\ \text{Spec}(V) & \longrightarrow & S \end{array}$$

any isomorphism between the restrictions of  $g_1$  and  $g_2$  to the generic point of  $\text{Spec}(V)$  can be extended to an isomorphism between  $g_1$  and  $g_2$ .

This criterion is nothing other but the valuative criterion of properness (EGA, II, 7.3.8) applied to the (representable) diagonal morphism.

**Theorem (4.19)** (Valuative criterion for properness.) — *If  $f$  is separated, then  $f$  is proper if and only if, for any discrete valuation ring  $V$  with field of fractions  $K$  and any commutative diagram*

$$\begin{array}{ccc} & & T \\ g \nearrow & & \downarrow f \\ \text{Spec}(K) & \longrightarrow & \text{Spec}(V) \longrightarrow S \end{array}$$

there exists a finite extension  $K'$  of  $K$  such that  $g$  extends to  $\text{Spec}(V')$ , where  $V'$  is the integral closure of  $V$  in  $K'$

$$\begin{array}{ccccc} & & T & & \\ & g \nearrow & \downarrow f & \nwarrow & \\ \text{Spec}(K') & \longrightarrow & \text{Spec}(V') & & \\ & \downarrow & \downarrow & & \\ \text{Spec}(K) & \longrightarrow & \text{Spec}(V) & \longrightarrow & S \end{array}$$

To prove a given  $f$  is proper, it suffices to verify the above criterion under the additional hypothesis that  $V$  is complete and has an algebraically closed residue field. Further, given a dense open subset  $U$  of  $T$ , it is enough to test only  $g$ 's which factor through  $U$ .

**Proposition (4.20).** — *Let  $\mathcal{S}$  be an algebraic stack. The functor which, to any algebraic stack over  $\mathcal{S}$ ,  $f: \mathcal{E} \rightarrow \mathcal{S}$ , associates the  $\mathcal{O}_{\mathcal{S}}$  sheaf of algebras  $f_* \mathcal{O}_{\mathcal{E}}$  induces an equivalence of categories between:*

- (i) the category of algebraic stacks representable and affine over  $\mathcal{S}$ ;  
 (ii) the dual of the category of quasi-coherent  $\mathcal{O}_{\mathcal{S}}$ -algebras.

Let  $\mathcal{A}$  be a quasi-coherent sheaf of  $\mathcal{O}_{\mathcal{S}}$ -algebras on an algebraic stack  $\mathcal{S}$ . For each étale morphism  $x: U \rightarrow \mathcal{S}$ , with  $U$  affine, let  $\mathcal{A}'(U)$  be the integral closure of  $\Gamma(U, \mathcal{O}_{\mathcal{S}})$  in  $\mathcal{A}(U)$ . By (EGA, II, 6.3.4), the  $\mathcal{A}'(U)$  for variable  $U$  are the sections over  $U$  of a quasi-coherent sheaf  $\mathcal{A}'$  on  $\mathcal{S}$ , which will be called the *integral closure* of  $\mathcal{O}_{\mathcal{S}}$  in  $\mathcal{A}$ .

Let  $f: \mathcal{E} \rightarrow \mathcal{S}$  be representable and affine. The algebraic stack which is associated by (4.20) to the integral closure of  $\mathcal{O}_{\mathcal{S}}$  in  $f_*\mathcal{O}_{\mathcal{E}}$  will be called the *normalization* of  $\mathcal{S}$  with respect to  $\mathcal{E}$ . Its formation is compatible with any étale change of basis.

**Theorem (4.21).** — *Let  $\mathcal{S}$  be a quasi-separated stack over a noetherian scheme  $S$ .*

*Assume that*

- (i) *the diagonal map  $\mathcal{S} \rightarrow \mathcal{S} \times_S \mathcal{S}$  is representable and unramified;*
- (ii) *there exists a scheme  $X$  of finite type over  $S$  and a smooth and surjective  $S$ -morphism from  $X$  to  $\mathcal{S}$ .*

*Then,  $\mathcal{S}$  is an algebraic stack of finite type over  $S$ .*

M. Artin has developed powerful methods to relate pro-representability of a stack to the existence of étale surjective maps  $x: X \rightarrow \mathcal{S}$ .

## § 5. Second proof of the irreducibility theorem.

Let  $\mathcal{M}_g$  ( $g \geq 2$ ) be the stack whose category of sections over a scheme  $S$  is the category of stable curves of genus  $g$  over  $S$ , the morphisms being the isomorphisms of schemes over  $S$ . By (1.11), the diagonal morphism  $\Delta: \mathcal{M}_g \rightarrow \mathcal{M}_g \times_S \mathcal{M}_g$  is representable, finite and unramified.

We saw in § 1 that the stack classifying the tricanonically embedded stable curves of genus  $g$  is represented by a scheme  $H_g$ , smooth and of finite type over  $\text{Spec}(\mathbb{Z})$ . The “forgetful” morphism

$$H_g \rightarrow \mathcal{M}_g$$

is representable, smooth and surjective. Indeed, if  $p: C \rightarrow S$  is a stable curve over  $S$ , defining a morphism

$$\gamma: S \rightarrow \mathcal{M}_g,$$

then the fibre product  $H_g \times_{\mathcal{M}_g} S$  is the scheme, smooth over  $S$ , of isomorphisms between the standard projective space of dimension  $5g-6$  over  $S$  and  $\mathbb{P}(p_*(\omega_{C/S}^{\otimes 3}))$ . We deduce from this and (4.21) that:

**Proposition (5.1).** —  *$\mathcal{M}_g$  is a separated algebraic stack of finite type over  $\text{Spec}(\mathbb{Z})$ .*

Let us denote by  $\mathcal{M}_g^0$  the open subset of  $\mathcal{M}_g$  which “consists of” smooth curves, and by  $\mathcal{X}_g$  the “universal curve” over  $\mathcal{M}_g$ , the algebraic stack classifying pointed stable curves.

**Theorem (5.2).** — *The algebraic stacks  $\mathcal{M}_g$  and  $\mathcal{X}_g$  are proper and smooth over  $\text{Spec}(\mathbb{Z})$  and the complement of  $\mathcal{M}_g^0$  in  $\mathcal{M}_g$  is a divisor with normal crossings relative to  $\text{Spec}(\mathbb{Z})$ .*

*Proof.* — Let  $x: X \rightarrow \mathcal{M}_g$  be étale and consider the following commutative diagram of algebraic stacks of finite type over  $\text{Spec}(\mathbf{Z})$ :

$$\begin{array}{ccccc}
 X \times_{\mathcal{M}_g} \mathcal{Z}_g & \xrightarrow{\quad} & \mathcal{Z}_g & \searrow & \\
 \downarrow & \searrow & \downarrow & & \downarrow q \\
 & X \times_{\mathcal{M}_g} H_g & \xrightarrow{x'} & H_g & \\
 \downarrow & \downarrow q' & & & \downarrow q \\
 X \times_{\mathcal{M}_g} \mathcal{Z}_g & \xrightarrow{\quad} & \mathcal{Z}_g & \searrow p & \\
 \downarrow & \searrow & \downarrow & & \downarrow \\
 X & \xrightarrow{x} & \mathcal{M}_g & & 
 \end{array}$$

In this diagram, the four horizontal arrows are étale and the four vertical arrows are smooth and surjective. As  $H_g$  and  $\mathcal{Z}_g$  (p. 78), are smooth over  $\text{Spec}(\mathbf{Z})$ , so are  $X$  and  $X \times_{\mathcal{M}_g} \mathcal{Z}_g$ . In addition,  $q'^{-1}x'^{-1}(\mathcal{M}_g^0) = x'^{-1}(H_g^0)$  is the complement of a divisor with normal crossings relative to  $\text{Spec}(\mathbf{Z})$ , so that the inclusion of  $x^{-1}(\mathcal{M}_g)$  in  $X$  has the same property. This being true for any  $x, \mathcal{M}_g$  and  $\mathcal{Z}_g$  are smooth over  $\text{Spec}(\mathbf{Z})$  and  $\mathcal{M}_g^0$  is the complement of a divisor with normal crossings relative to  $\text{Spec}(\mathbf{Z})$ . In particular  $\mathcal{M}_g^0$  is dense in  $\mathcal{M}_g$ .

We may now use the valuative criterion for properness in its modified form (4.19) to deduce the properness of  $\mathcal{M}_g$  from the stable reduction theorem. The properness of  $\mathcal{Z}_g$  then results from that of  $p$ .

(5.3) Let  $p: X \rightarrow S$  be a stable smooth curve of genus  $g \geq 2$  over  $S$ . If  $k \in \mathbf{N}$  is invertible in  $\mathcal{O}_S$ , the sheaf  $R^1 p_* (\mathbf{Z}/k\mathbf{Z})$  on  $S_{\text{ét}}$  is locally free over  $\mathbf{Z}/k\mathbf{Z}$ , of rank  $2g$ , and the cup-product is a non degenerate alternating form

$$R^1 p_* (\mathbf{Z}/k\mathbf{Z}) \otimes R^1 p_* (\mathbf{Z}/k\mathbf{Z}) \rightarrow R^2 p_* (\mathbf{Z}/k\mathbf{Z}) \sim \mu_k^{\otimes -1}.$$

Locally on  $S_{\text{ét}}$ ,  $\mu_k$  is isomorphic to  $\mathbf{Z}/k\mathbf{Z}$ , and thus, locally,  $R^1 p_* (\mathbf{Z}/k\mathbf{Z})$  is provided with a non degenerate symplectic form with values in  $\mathbf{Z}/k\mathbf{Z}$ , which is determined up to a unit in  $\mathbf{Z}/k\mathbf{Z}$ , or, as we shall say,  $R^1 p_* (\mathbf{Z}/k\mathbf{Z})$  is provided with an homogeneous symplectic structure. The constant sheaf  $(\mathbf{Z}/k\mathbf{Z})^{2g}$  will be provided with the homogeneous symplectic structure induced by the standard symplectic structure of  $(\mathbf{Z}/k\mathbf{Z})^{2g}$ .

*Definition (5.4).* — A Jacobi structure of level  $k$  on  $X$  is an isomorphism (respecting their homogeneous symplectic structures) between  $R^1 p_* (\mathbf{Z}/k\mathbf{Z})$  and  $(\mathbf{Z}/k\mathbf{Z})^{2g}$ .

(5.5) Given a section  $s$  of  $p$ , and a set of prime numbers  $\mathbf{P}$  including all residue characteristics of  $S$ , the specialisation theorem for the fundamental group enables one to construct a pro-object  $\pi_1(X/S, s)^{(\mathbf{P})}$  of the category (l.c.  $\text{gr}^{(\mathbf{P})}$ ) of locally constant sheaves of finite groups of order prime to  $\mathbf{P}$ , with the properties

(i)  $\text{Hom}_s(\pi_1(X/S, s)^{(\mathbf{P})}, G) = R^1 p_* (X \bmod s, p^* G) = R^1 p_* (\text{Ker}(p^* G \rightarrow s_* G))$  functorially in  $G \in (\text{l.c. } \text{gr}^{(\mathbf{P})})$ ;

(ii) the formation of  $\pi_1(X/S, s)^{(\mathbf{P})}$  is compatible with any change of base.

If  $G$  and  $H$  are two sheaves of groups on  $S_{\text{ét}}$ , we define the sheaf of exterior morphisms from  $H$  to  $G$ ,  $\text{Hom}^{\text{ext}}(H, G)$ , as the quotient of  $\text{Hom}(H, G)$  by the action of  $H$  induced by its action on itself by inner automorphism.

The sheaf  $\text{Hom}_S^{\text{ext}}(\pi_1(X/S, s)^{(\mathbb{P})}, G) \sim R^1 p_* (p^* G)$  (for  $G \in (\text{I.c. gr}^{(\mathbb{P})})$ )

is “independent” of the choice of  $s$ .

We shall denote it as

$$\text{Hom}_S^{\text{ext}}(\pi_1(X/S)^{(\mathbb{P})}, G).$$

As  $p: X \rightarrow S$  admits sections locally for the étale topology, this sheaf makes sense without assuming  $p$  to have a global section. It is independent of the choice of  $\mathbb{P}$ , so long as  $\mathbb{P}$  is prime to the order of  $G$  and includes all residue characteristics of  $S$ .

**Definition (5.6).** — Let  $G$  be a finite group of order  $n$  prime to  $\mathbb{P}$ . A Teichmüller structure of level  $G$  on  $X$  is a surjective exterior homomorphism from  $\pi_1(X/S)$  to  $G$ .

The finite generation of  $\pi_1(X/S)$  (SGA, 60/61, exp. 10) implies:

**Lemma (5.7).** — The sheaf on  $(\text{Sch}/S)$  of the Teichmüller structures of level  $G$  on  $X$  is represented by an étale covering of  $S$ .

We denote by  ${}_G\mathcal{M}_g^0$  the stack classifying the stable smooth curves of genus  $g$  and characteristic prime to  $n$ , with a Teichmüller structure of level  $G$ .

For any algebraic stack  $\mathcal{M}$ , we denote by  $\mathcal{M}[1/n]$  its open subset  $\mathcal{M} \times \text{Spec}(\mathbb{Z}[1/n])$ .

**Lemma (5.7)** may now be rephrased:

**Proposition (5.8).** — The “forgetful” morphism

$${}_G\mathcal{M}_g^0 \rightarrow \mathcal{M}_g^0[1/n]$$

is representable, finite and étale.

The stack  ${}_G\mathcal{M}_g^0$  thus is an algebraic stack. Let  ${}_G\mathcal{M}_g$  be the normalisation of  $\mathcal{M}_g[1/n]$  with respect to  ${}_G\mathcal{M}_g^0$ . The stack  ${}_G\mathcal{M}_g$ , being representable and finite over  $\mathcal{M}_g[1/n]$ , is proper over  $\text{Spec}(\mathbb{Z}[1/n])$ .

**Theorem (5.9).** — The geometric fibres of the projection of  ${}_G\mathcal{M}_g$  onto  $\text{Spec}(\mathbb{Z}[1/n])$  are normal, and, fibre by fibre,  ${}_G\mathcal{M}_g^0$  is dense in  ${}_G\mathcal{M}_g$ .

*Proof.* — We will use Abhyankar-Artin’s lemma, in its “absolute” form:

**Lemma (5.10).** — Let  $D$  be a divisor with normal crossings on an excellent regular scheme  $X$ ,  $Y$  an étale covering of  $X - D$  and  $\bar{Y}$  the normalisation of  $X$  with respect to  $Y$ . Assume that the generic points of the irreducible components of  $D$  are all of characteristic 0. Then every geometric point of  $X$  has an étale neighbourhood  $x: X' \rightarrow X$  such that, on  $X'$ :

- (i)  $D$  becomes a union of regular divisors  $(D_i)_{i \in I}$ ,  $D_i$  of equation  $t_i = 0$ .
- (ii) There exists an integer  $k$  prime to residue characteristics of  $X'$  such that  $\bar{Y}$  becomes isomorphic to a disjoint union of quotients (by subgroups of  $\mu_k^1$ ) of the covering of  $X'$  obtained by extracting the  $k^{\text{th}}$ -roots of the  $t_i$ ’s.

If  $x: X \rightarrow \mathcal{M}_g \left[ \frac{1}{n} \right]$  is an étale morphism, then  $X^0 = x^{-1}(\mathcal{M}_g^0)$  is the complement in  $X$  of a divisor with normal crossing relative to  $\text{Spec}(\mathbb{Z})$ ,  $X_1^0 = X \times_{\mathcal{M}_g} ({}_G\mathcal{M}_g^0)$  is an étale

covering of  $X^0$  and  $X_1 = S \times_{\mathcal{M}_g} ({}_G\mathcal{M}_g)$  is the normalisation of  $X$  with respect to this covering of  $X^0$ .

By the explicit local description (5.10), for any prime number  $l$  prime to  $n$ ,  $X_1 \times \text{Spec}(\overline{\mathbb{F}}_l)$  is the normalisation of  $X \times \text{Spec}(\overline{\mathbb{F}}_l)$  with respect to  $X_1 \times \text{Spec}(\overline{\mathbb{F}}_l)$ . As this is true for any modular family, we get (5.9).

*Corollary (5.11). — The geometric fibres of the projection*

$${}_G\mathcal{M}_g^0 \rightarrow \text{Spec}\left(\mathbb{Z}\left[\frac{1}{n}\right]\right)$$

*all have the same number of connected components.*

*Proof.* — By (4.17), all geometric fibres  ${}_G\mathcal{M}_g \times \text{Spec}(\overline{\mathbb{F}}_l)$  of  ${}_G\mathcal{M}_g$  over  $\text{Spec}(\mathbb{Z}[1/n])$  have the same number of connected components. These connected components are irreducible (4.16). Furthermore  ${}_G\mathcal{M}_g^0 \times \text{Spec}(\overline{\mathbb{F}}_l)$  is dense in  ${}_G\mathcal{M}_g \times \text{Spec}(\overline{\mathbb{F}}_l)$  and thus has the same number of connected (or irreducible) components as  ${}_G\mathcal{M}_g \times \text{Spec}(\overline{\mathbb{F}}_l)$ , and this number is independent of  $l$ .

(5.12) Let us denote by  $\Pi$  the fundamental group of an oriented closed differentiable surface  $S_0$  of genus  $g$ . The group  $\Pi$  may be defined by generators and relations as follows:

- (i) generators:  $x_i$  for  $1 \leq i \leq 2g$ ;
- (ii) relation:  $(x_1, x_{g+1}) \dots (x_i, x_{g+i}) \dots (x_g, x_{2g}) = e$ , where  $(a, b) = aba^{-1}b^{-1}$ .

Let  $(e_i)$  be the standard basis of  $\mathbb{Z}^{2g}$ . The morphism  $\varphi$  from  $\Pi$  to  $\mathbb{Z}^{2g}$  with  $\varphi(X_i) = e_i$  identifies  $\Pi/(\Pi, \Pi) = H_1(\Pi, \mathbb{Z})$  with  $\mathbb{Z}^{2g}$ .

The surface  $S_0$  is a  $K(\Pi, 1)$  and thus

$$H^2(\Pi, \mathbb{Z}) = H^2(S_0, \mathbb{Z}) = \mathbb{Z},$$

and the cup product defines a symplectic structure on  $\Pi/(\Pi, \Pi)$ . This structure is identified by  $\varphi$  with the standard symplectic structure of  $\mathbb{Z}^{2g}$ .

We denote by  $\text{Aut}^0(\Pi)$  the subgroup of  $\text{Aut}(\Pi)$  which acts trivially on  $H^2(\Pi, \mathbb{Z})$ . Dehn has proved that each exterior automorphism of  $\Pi$  is induced by a diffeomorphism of  $S_0$  onto itself and that the map induced by  $\varphi$ :

$$\text{Aut}^0(\Pi) \rightarrow \mathbf{Sp}_{2g}(\mathbb{Z}),$$

is surjective (see [Ma]).

*Theorem (5.13). — The number of connected components of any geometric fibre of the projection of  ${}_G\mathcal{M}_g^0$  onto  $\text{Spec}(\mathbb{Z}[1/n])$  is equal to the number of orbits of  $\text{Aut}^0(\Pi)$  in the set of exterior epimorphisms from  $\Pi$  to  $G$ .*

*Proof.* — By (5.13), it suffices to prove that  ${}_G\mathcal{M}_g^0 \times \text{Spec}(\mathbb{C})$  has the said number of connected components. As results from (4.14),

$$\pi_0({}_G\mathcal{M}_g^0 \times \text{Spec}(\mathbb{C})) = \pi_0({}_G\mathcal{M}_g)$$

where  ${}_G\mathcal{M}_g$  is the coarse moduli scheme classifying stable smooth curves of genus  $g$  over  $\mathbb{C}$  with a Teichmüller structure of level  $G$ .



Recall that a *Teichmüller curve* of genus  $g$  is a stable smooth curve  $C$  of genus  $g$  over  $\mathbb{C}$  together with an exterior isomorphism  $\varphi$  of the transcendental fundamental group  $\pi_1(C)$  with  $\Pi$ , which induces a symplectic isomorphism <sup>(1)</sup> between

$$H_1(C, \mathbb{Z}) \sim \pi_1(C) / (\pi_1(C), \pi_1(C))$$

and  $\Pi/(\Pi, \Pi)$ . By Teichmüller's theory [W], the analytic space  $T_g$  classifying Teichmüller curves of a given genus  $g \geq 2$  is homeomorphic to a ball, and hence connected.

If  $\psi$  is a surjective homomorphism from  $\Pi$  to  $G$ , the map

$$(C, \varphi) \mapsto (C, \psi\varphi)$$

defines a morphism  $t_\psi: T_g \rightarrow {}_G M_g$ . Two such maps  $t_\psi$  and  $t_{\psi'}$  have the same image if and only if  $\psi = \psi'\sigma$  for  $\sigma \in \text{Aut}^0(\Pi)$ , and

$${}_G M_g = \coprod_{\psi \bmod \text{Aut}^0(\Pi)} t_\psi(T_g)$$

which implies (5.13).

(5.14) Let us denote by  ${}_n \mathcal{M}_g^0$  the algebraic stack classifying stable smooth curves with a Jacobi structure of level  $n$ . This algebraic stack "is" a true scheme for  $n \geq 3$  (by [S]).

If  $\varphi$  is a Jacobi structure of level  $n$  on a stable smooth curve  $p: X \rightarrow S$ , we define the "multiplier"  $\mu(\varphi)$  of  $\varphi$  by the commutative diagram

$$\begin{array}{ccc} \wedge^2(\mathbb{Z}/n\mathbb{Z})^{2g} & \longrightarrow & \mathbb{Z}/n\mathbb{Z} \\ \downarrow \wedge^2 \varphi & & \downarrow \mu(\varphi) \\ \wedge^2 R^1 p_* (\mathbb{Z}/n\mathbb{Z}) & \xrightarrow{\wedge} & \mu_n^{\otimes -1} \sim R^2 p_* (\mathbb{Z}/n\mathbb{Z}) \end{array}$$

The scheme of isomorphisms between  $\mathbb{Z}/n\mathbb{Z}$  and  $\mu_n^{\otimes -1}$  is  $\text{Spec}(\mathbb{Z}[e^{2\pi i/n}, 1/n])$  thus  $\varphi \mapsto \mu(\varphi)$  defines a morphism  $\mu$  from  $S$  to  $\text{Spec}(\mathbb{Z}[e^{2\pi i/n}, 1/n])$ . This being true for any  $X$  and  $S$ ,  $\mu$  is induced by

$$\mu: {}_n \mathcal{M}_g^0 \rightarrow \text{Spec}(\mathbb{Z}[e^{2\pi i/n}, 1/n]).$$

*Theorem (5.15).* — *The geometric fibres of the morphism  $\mu$  are connected.*

*Proof.* — By definition,  ${}_n \mathcal{M}_g^0$  is open and closed in  ${}_G \mathcal{M}_g^0$  for  $G = (\mathbb{Z}/n\mathbb{Z})^{2g}$ . The group  $\text{GL}_{2g}(\mathbb{Z}/n\mathbb{Z})$  acts on  $G$ , and thus on  ${}_G \mathcal{M}_g^0$ . One has:

(i) the open subset  ${}_n \mathcal{M}_g^0$  of  ${}_G \mathcal{M}_g^0$  is stable under the subgroup  $H = \text{CSp}_{2g}(\mathbb{Z}/n\mathbb{Z})$  of symplectic similitudes;

(ii)  ${}_G \mathcal{M}_g^0 = \coprod_{G/H} \sigma({}_n \mathcal{M}_g^0)$ .

<sup>(1)</sup> An arbitrary isomorphism  $\varphi$  induces an isomorphism between  $H_1(C, \mathbb{Z})$  and  $\Pi/(\Pi, \Pi)$  which is always symplectic up to sign.

It now results from (5.11) that all geometric fibres of  $\mu$  have the same number of connected components.

Consider the geometric fibre of  $\mu$  at the standard complex place of  $\text{Spec}(\mathbf{Z}[e^{2\pi i/n}, 1/n])$ . This fibre is the algebraic stack classifying complex stable smooth curves  $C$  provided with a symplectic isomorphism

$$H^1(C, \mathbf{Z}/n\mathbf{Z}) \xrightarrow{\sim} (\mathbf{Z}/n\mathbf{Z})^g.$$

Reasoning as in (5.13), we are reduced to proving

**Lemma (5.16).** — *The homomorphism*

$$\text{Aut}^0(\Pi) \rightarrow \mathbf{Sp}_{2g}(\mathbf{Z}/n\mathbf{Z})$$

*is surjective.*

*Proof.* — This results from Dehn's theory (5.12) and from the fact that the groups  $\mathbf{Sp}_n$ , being split semi-simple simply connected groups, are generated by their unipotent elements, so that the reduction map

$$\mathbf{Sp}_{2g}(\mathbf{Z}) \rightarrow \mathbf{Sp}_{2g}(\mathbf{Z}/n\mathbf{Z})$$

is surjective.

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